Automated Parcel Damage Detection System for E-Commerce Fulfillment Centres

Model Training & Testing

Overview

The model training and testing phase focuses on developing an accurate parcel damage detection system using deep learning. RF-DETR Small was trained on an augmented dataset to improve detection performance and support real-time deployment in e-commerce fulfillment centers.

Key Points

Model: RF-DETR Small
Dataset Size: 3519 Images
Training Epochs: 165
Application: Parcel Damage Detection
Supports Real-Time Inspection

Training Results

Training Performance

The model was trained for 165 epochs on the Roboflow cloud platform. During training, all loss curves decreased steadily, indicating successful learning and convergence

Observations

Box Loss reduced significantly.

Classification Loss improved consistently.

Object Detection Loss converged successfully.

Model learned to localize and classify damages effectively.

Model Performance

mAP Performance Analysis

The model showed rapid learning during the initial training stages and achieved stable performance after 100 epochs.

Key Results

Fast learning in early epochs.

Peak performance reached around 72–73%.

Stable convergence achieved.

Minor overfitting observed near the final epoch.

Final Model Metrics
Validation Results

The trained model achieved strong detection performance on the validation dataset.

Metric	Value
Overall mAP@50	73%
Severity 1 Accuracy	71%
Severity 2 Accuracy	75%

Observation: The model detects severe parcel damage more accurately than minor damage.

Average Precision by Class
Per-Class Performance

The model performed well across different severity categories.

Results

Overall Accuracy: 73%

Severity 1 Damage: 71%

Severity 2 Damage: 75%

Analysis

Severe damages are easier to detect.

Additional training data can further improve minor damage detection.

Testing Observations

Testing Results

During manual testing, some images produced no detections due to differences from training data.

Reasons

Different lighting conditions.
High confidence threshold settings.
Non-standard parcel images.

Achievements

Successful model convergence.
Improved accuracy through augmentation.
Reliable severity classification.

Conclusion

The RF-DETR Small model successfully achieved 73% detection accuracy for parcel damage identification. The training results demonstrate the effectiveness of deep learning for automated quality inspection. With further dataset expansion and real-time deployment, the system can significantly improve parcel inspection efficiency in e-commerce fulfillment centers.

THANK YOU